NAMA : Marselia Yura Alinski

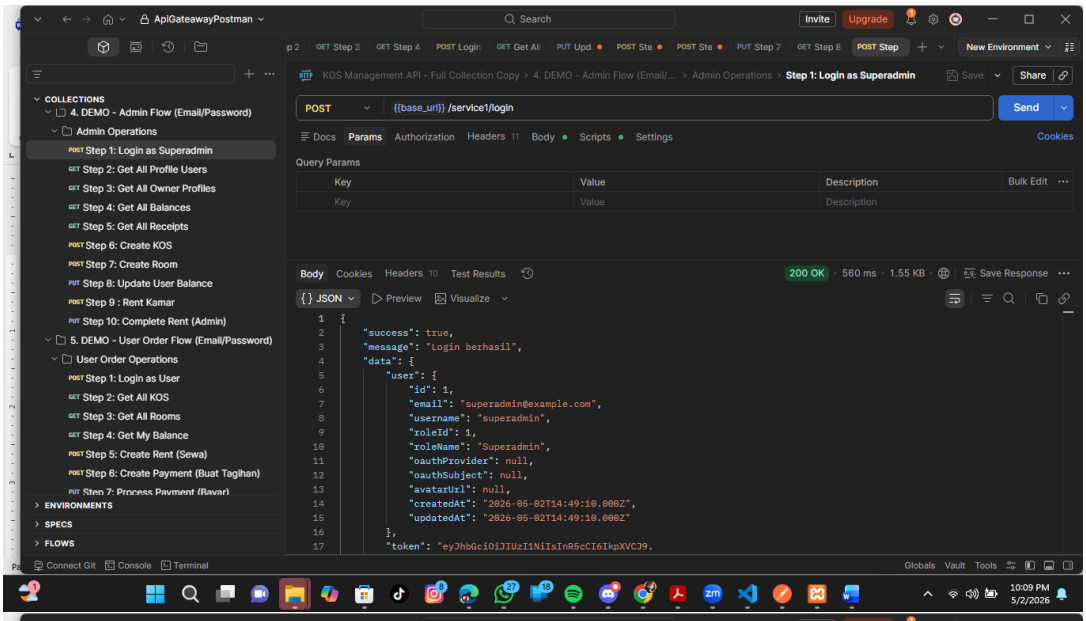
NIM : 2410511008

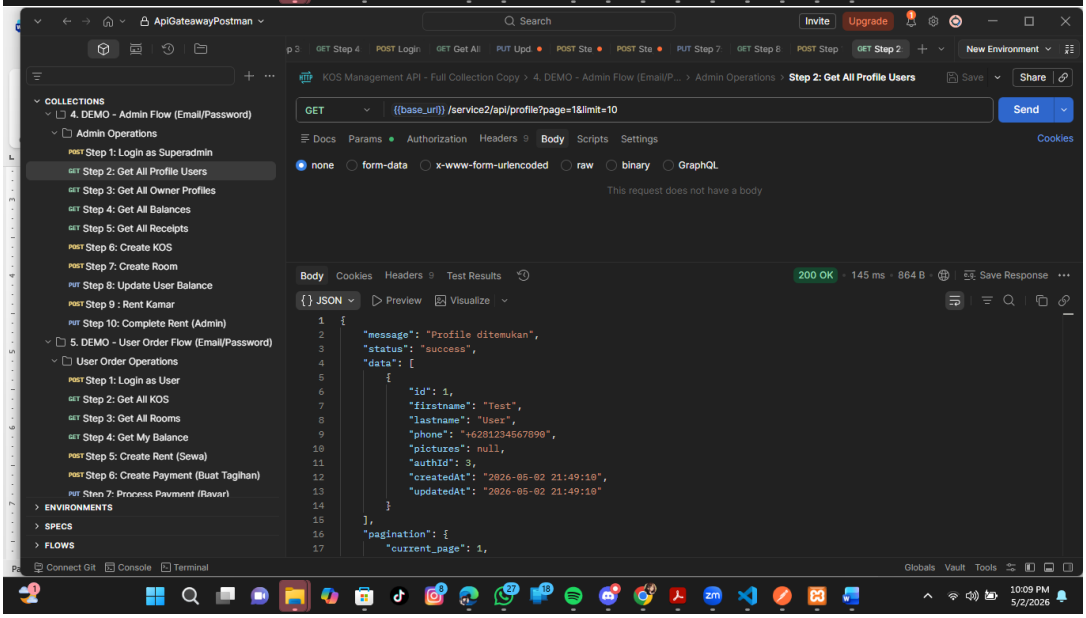
KELAS : A

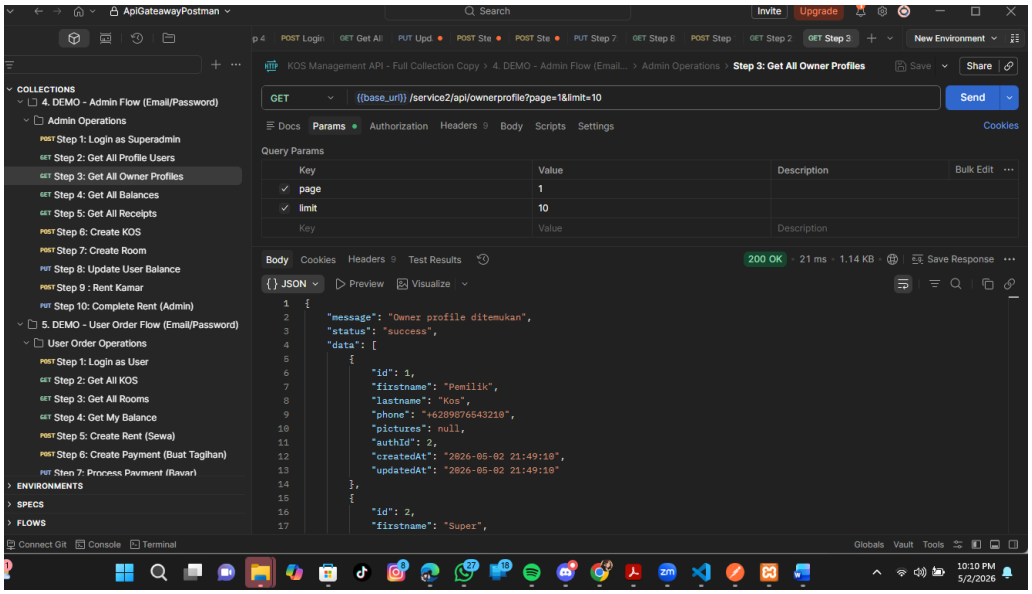
UTS “SISTEM MANAJEMEN KOS”

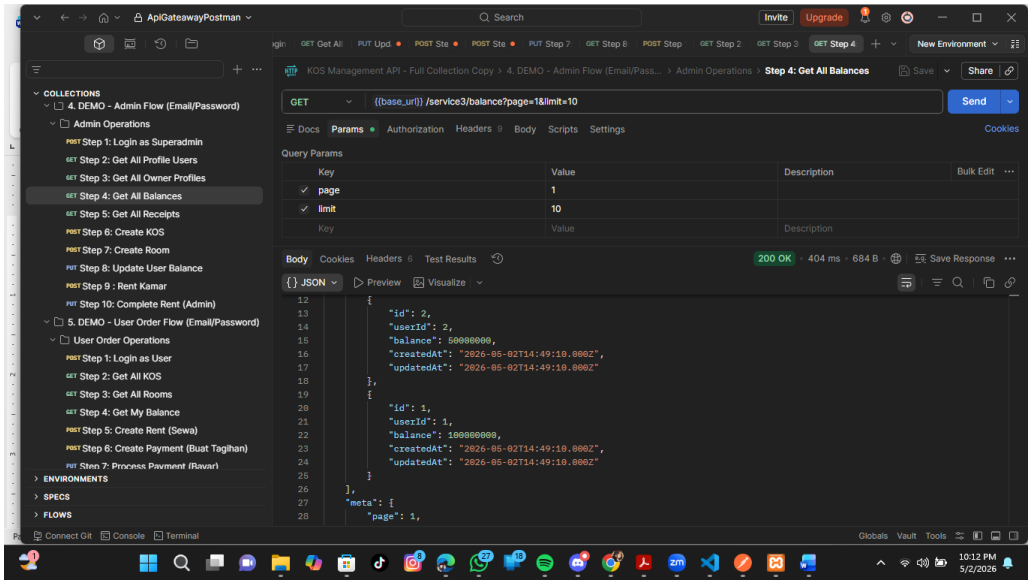
Screnshooth postman

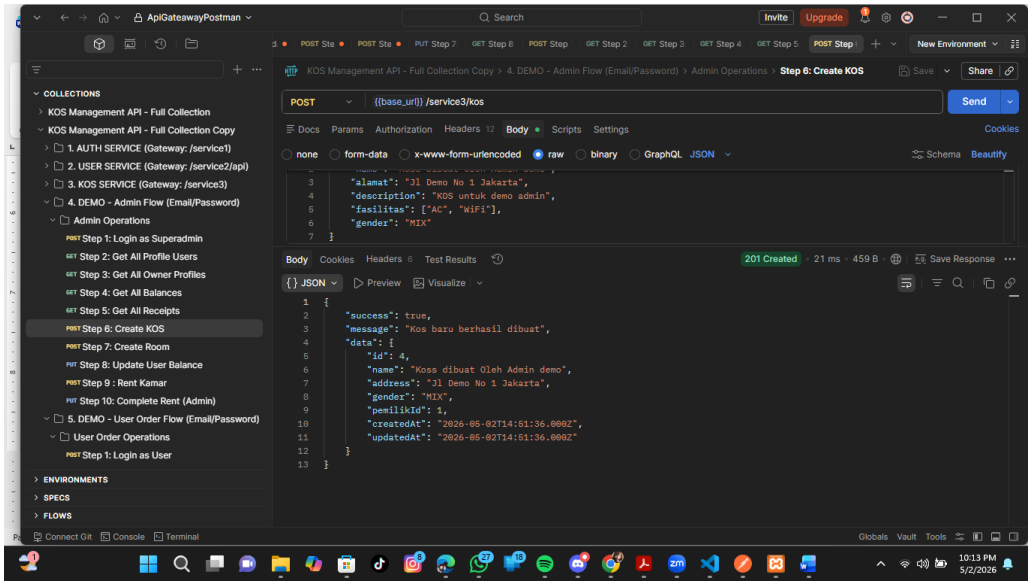
1) Admin opration

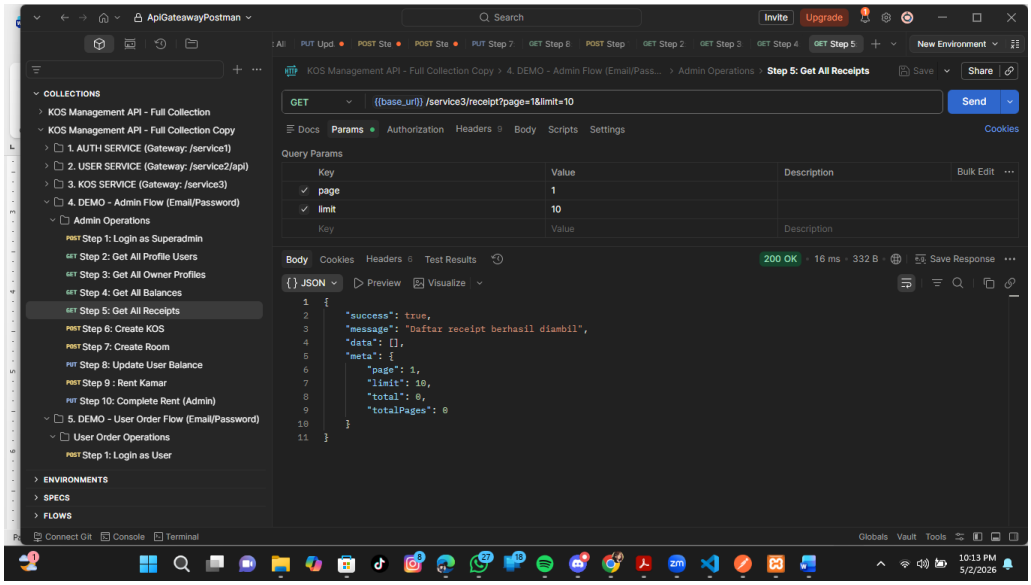





ApiGateawayPostman
Search
Invite
Upgrade
New Environment
COLLECTIONS
4. DEMO - Admin Flow (Email/Password)
Admin Operations
Step 1: Login as Superadmin
Step 2: Get All Profile Users
Step 3: Get All Owner Profiles
Step 4: Get All Balances
Step 5: Get All Receipts
Step 6: Create KOS
Step 7: Create Room
Step 8: Update User Balance
Step 9 : Rent Kamar
Step 10: Complete Rent (Admin)
5. DEMO - User Order Flow (Email/Password)
User Order Operations
Step 1: Login as User
Step 2: Get All KOS
Step 3: Get All Rooms
Step 4: Get My Balance
Step 5: Create Rent (Sewa)
Step 6: Create Payment (Buat Tagihan)
ENVIRONMENTS
SPECS
FLOWS
KOS Management API - Full Collection Copy > 4. DEMO - Admin Flow (Email... > Admin Operations > Step 3: Get All Owner Profiles
Save
Share
GET
{{base_url}} /service2/api/ownerprofile?page=1&limit=10
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
page
1
limit
10
Body
Cookies
Headers 9
Test Results
200 OK
21 ms
1.14 KB
Save Response
JSON
Preview
Visualize
{
"message": "Owner profile ditemukan",
"status": "success",
"data": [
{
"id": 1,
"firstname": "Pemilik",
"lastname": "Kos",
"phone": "+6289876543210",
"pictures": null,
"authId": 2,
"createdAt": "2026-05-02 21:49:10",
"updatedAt": "2026-05-02 21:49:10"
},
{
"id": 2,
"firstname": "Super",
Connect Git
Console
Terminal
Globals
Vault
Tools
10:10 PM
5/2/2026


ApiGateawayPostman
Search
Invite
Upgrade
New Environment
COLLECTIONS
4. DEMO - Admin Flow (Email/Password)
Admin Operations
Step 1: Login as Superadmin
Step 2: Get All Profile Users
Step 3: Get All Owner Profiles
Step 4: Get All Balances
Step 5: Get All Receipts
Step 6: Create KOS
Step 7: Create Room
Step 8: Update User Balance
Step 9 : Rent Kamar
Step 10: Complete Rent (Admin)
5. DEMO - User Order Flow (Email/Password)
User Order Operations
Step 1: Login as User
Step 2: Get All KOS
Step 3: Get All Rooms
Step 4: Get My Balance
Step 5: Create Rent (Sewa)
Step 6: Create Payment (Buat Tagihan)
ENVIRONMENTS
SPECS
FLOWS
KOS Management API - Full Collection Copy > 4. DEMO - Admin Flow (Email/Pass... > Admin Operations > Step 4: Get All Balances
Save
Share
GET
{{base_url}} /service3/balance?page=1&limit=10
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
page
1
limit
10
Body
Cookies
Headers 6
Test Results
200 OK
404 ms
684 B
Save Response
JSON
Preview
Visualize
"id": 2,
"userId": 2,
"balance": 50000000,
"createdAt": "2026-05-02T14:49:10.000Z",
"updatedAt": "2026-05-02T14:49:10.000Z"
},
"id": 1,
"userId": 1,
"balance": 100000000,
"createdAt": "2026-05-02T14:49:10.000Z",
"updatedAt": "2026-05-02T14:49:10.000Z"
],
"meta": {
"page": 1,
Connect Git
Console
Terminal
Globals
Vault
Tools
10:12 PM
5/2/2026


ApiGateawayPostman
Search
Invite
Upgrade
New Environment
KOS Management API - Full Collection Copy › 4. DEMO - Admin Flow (Email/Password) › Admin Operations › Step 6: Create KOS
Save
Share
COLLECTIONS
KOS Management API - Full Collection
KOS Management API - Full Collection Copy
1. AUTH SERVICE (Gateway: /service1)
2. USER SERVICE (Gateway: /service2/api)
3. KOS SERVICE (Gateway: /service3)
4. DEMO - Admin Flow (Email/Password)
Admin Operations
Step 1: Login as Superadmin
Step 2: Get All Profile Users
Step 3: Get All Owner Profiles
Step 4: Get All Balances
Step 5: Get All Receipts
Step 6: Create KOS
Step 7: Create Room
Step 8: Update User Balance
Step 9 : Rent Kamar
Step 10: Complete Rent (Admin)
5. DEMO - User Order Flow (Email/Password)
User Order Operations
Step 1: Login as User
ENVIRONMENTS
SPECS
FLOWS
POST
{{base_url}} /service3/kos
Send
Docs
Params
Authorization
Headers 12
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
"alamat": "Jl Demo No 1 Jakarta",
"description": "KOS untuk demo admin",
"fasilitas": ["AC", "WiFi"],
"gender": "MIX"
Body
Cookies
Headers 6
Test Results
201 Created
21 ms
459 B
Save Response
JSON
Preview
Visualize
"success": true,
"message": "Kos baru berhasil dibuat",
"data": {
"id": 4,
"name": "Koss dibuat Oleh Admin demo",
"address": "Jl Demo No 1 Jakarta",
"gender": "MIX",
"pemilikId": 1,
"createdAt": "2026-05-02T14:51:36.000Z",
"updatedAt": "2026-05-02T14:51:36.000Z"
Connect Git
Console
Terminal
Globals
Vault
Tools
10:13 PM
5/2/2026


ApiGateawayPostman
Search
Invite
Upgrade
New Environment
KOS Management API - Full Collection Copy › 4. DEMO - Admin Flow (Email/Pass... › Admin Operations › Step 5: Get All Receipts
Save
Share
COLLECTIONS
KOS Management API - Full Collection
KOS Management API - Full Collection Copy
1. AUTH SERVICE (Gateway: /service1)
2. USER SERVICE (Gateway: /service2/api)
3. KOS SERVICE (Gateway: /service3)
4. DEMO - Admin Flow (Email/Password)
Admin Operations
Step 1: Login as Superadmin
Step 2: Get All Profile Users
Step 3: Get All Owner Profiles
Step 4: Get All Balances
Step 5: Get All Receipts
Step 6: Create KOS
Step 7: Create Room
Step 8: Update User Balance
Step 9 : Rent Kamar
Step 10: Complete Rent (Admin)
5. DEMO - User Order Flow (Email/Password)
User Order Operations
Step 1: Login as User
ENVIRONMENTS
SPECS
FLOWS
GET
{{base_url}} /service3/receipt?page=1&limit=10
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
page
1
limit
10
Body
Cookies
Headers 6
Test Results
200 OK
16 ms
332 B
Save Response
JSON
Preview
Visualize
"success": true,
"message": "Daftar receipt berhasil diambil",
"data": [],
"meta": {
"page": 1,
"limit": 10,
"total": 0,
"totalPages": 0
Connect Git
Console
Terminal
Globals
Vault
Tools
10:13 PM
5/2/2026

KOS Management API - Full Collection Copy › 4. DEMO - Admin Flow (Email/Password) › Admin Operations › **Step 7: Create Room**

POST {{base_url}} /service3/room

Body: raw JSON

```
{
    "kosId": 4,
    "name": "Room A1",
    "description": "Kamar standar",
    "price": 100000,
```

201 Created · 17 ms · 438 B

```
{
    "success": true,
    "message": "Room baru berhasil dibuat",
    "data": {
        "id": 14,
        "name": "Room A1",
        "price": 100000,
        "capacity": 1,
        "status": "AVAILABLE",
        "kosId": 4,
        "createdAt": "2026-05-02T14:52:05.000Z",
        "updatedAt": "2026-05-02T14:52:05.000Z"
    }
}
```

KOS Management API - Full Collection Copy › 4. DEMO - Admin Flow (Email/...) › Admin Operations › **Step 8: Update User Balance**

PUT {{base_url}} /service3/balance/user/3

Body: raw JSON

```
{
    "balance": 50000000
}
```

200 OK · 16 ms · 391 B

```
{
    "success": true,
    "message": "Balance user berhasil diupdate",
    "data": {
        "id": 3,
        "userId": 3,
        "balance": 50000000,
        "createdAt": "2026-05-02T14:49:10.000Z",
        "updatedAt": "2026-05-02T14:52:23.000Z"
    }
}
```

KOS Management API - Full Collection Copy › 4. DEMO - Admin Flow (Email/Password) › Admin Operations › **Step 9 : Rent Kamar**

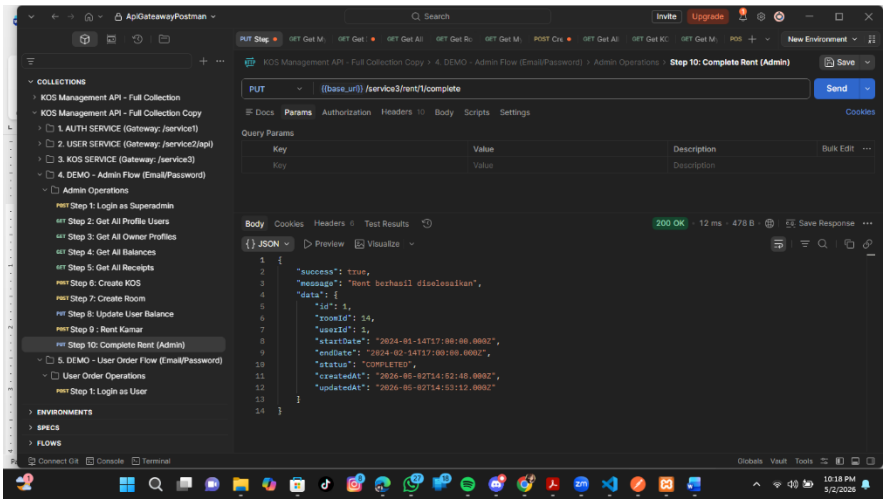
POST {{base_url}} /service3/rent

Body: raw JSON

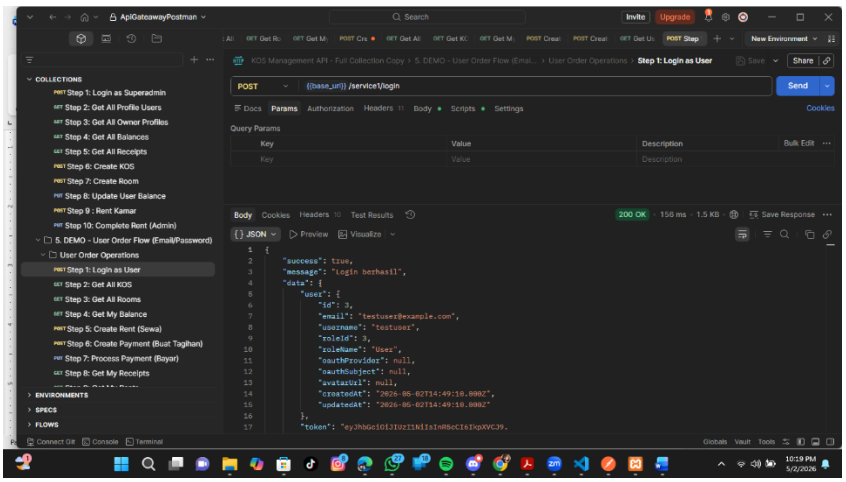
```
{
    "roomId":14,
    "startDate": "2024-01-15",
    "endDate": "2024-02-15"
}
```

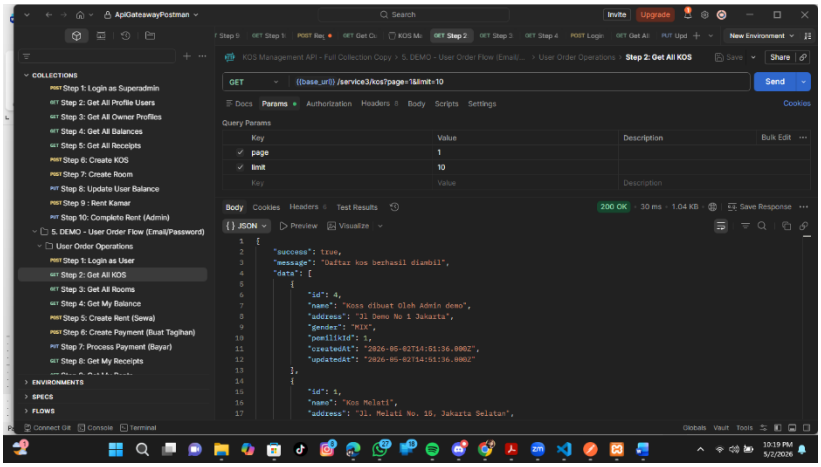
201 Created · 60 ms · 474 B

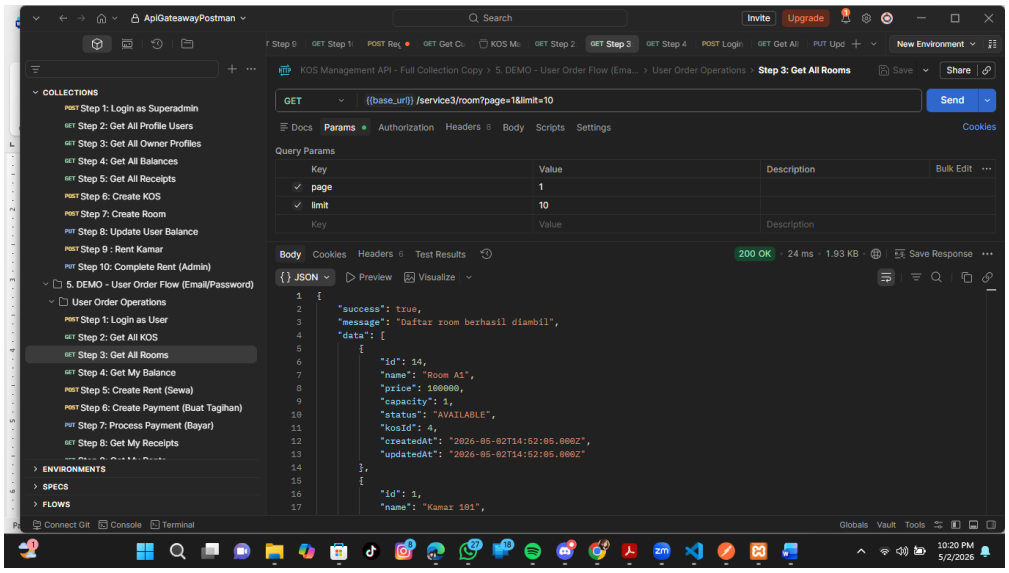
```
{
    "success": true,
    "message": "Rent berhasil dibuat",
    "data": {
        "id": 1,
        "roomId": 14,
        "userId": 1,
        "startDate": "2024-01-14T17:00:00.000Z",
        "endDate": "2024-02-14T17:00:00.000Z",
        "status": "ACTIVE",
        "createdAt": "2026-05-02T14:52:48.000Z",
        "updatedAt": "2026-05-02T14:52:48.000Z"
    }
}
```

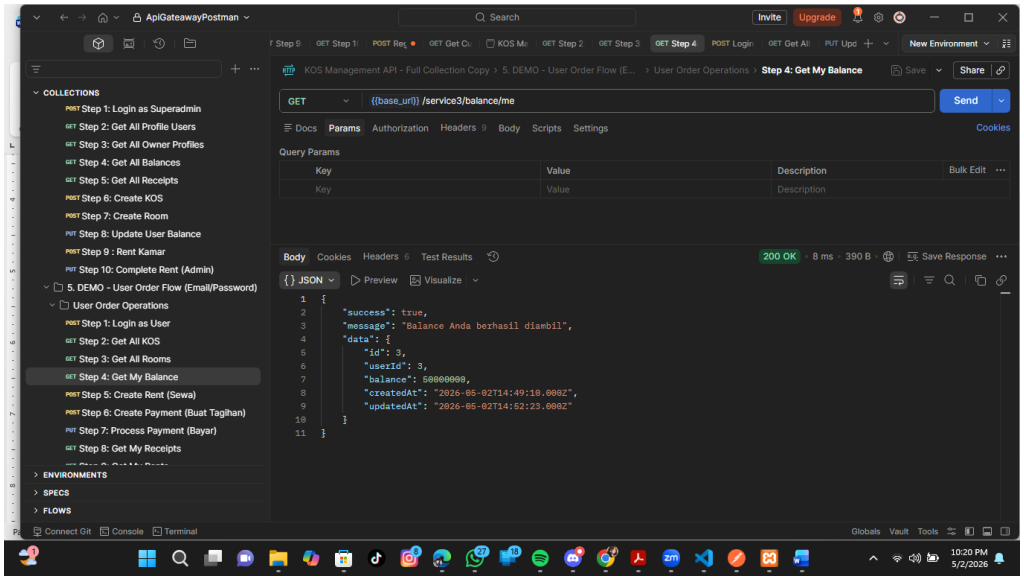


2) User order oprasion






ApiGateawayPostman
Search
Invite
Upgrade
New Environment
KOS Management API - Full Collection Copy > 5. DEMO - User Order Flow (Ema... > User Order Operations > Step 3: Get All Rooms
Save
Share
COLLECTIONS
POST Step 1: Login as Superadmin
GET Step 2: Get All Profile Users
GET Step 3: Get All Owner Profiles
GET Step 4: Get All Balances
GET Step 5: Get All Receipts
POST Step 6: Create KOS
POST Step 7: Create Room
PUT Step 8: Update User Balance
POST Step 9 : Rent Kamar
PUT Step 10: Complete Rent (Admin)
5. DEMO - User Order Flow (Email/Password)
User Order Operations
POST Step 1: Login as User
GET Step 2: Get All KOS
GET Step 3: Get All Rooms
GET Step 4: Get My Balance
POST Step 5: Create Rent (Sewa)
POST Step 6: Create Payment (Buat Tagihan)
PUT Step 7: Process Payment (Bayar)
GET Step 8: Get My Receipts
ENVIRONMENTS
SPECS
FLOWS
GET
{{base_url}} /service3/room?page=1&limit=10
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
page
1
limit
10
Body
Cookies
Headers 6
Test Results
200 OK
24 ms
1.93 KB
Save Response
JSON
Preview
Visualize
{
"success": true,
"message": "Daftar room berhasil diambil",
"data": [
{
"id": 14,
"name": "Room A1",
"price": 100000,
"capacity": 1,
"status": "AVAILABLE",
"kosId": 4,
"createdAt": "2026-05-02T14:52:05.000Z",
"updatedAt": "2026-05-02T14:52:05.000Z"
},
{
"id": 1,
"name": "Kamar 101",
Connect Git
Console
Terminal
Globals
Vault
Tools
10:20 PM
5/2/2026


ApiGateawayPostman
Search
Invite
Upgrade
New Environment
KOS Management API - Full Collection Copy > 5. DEMO - User Order Flow (E... > User Order Operations > Step 4: Get My Balance
Save
Share
COLLECTIONS
POST Step 1: Login as Superadmin
GET Step 2: Get All Profile Users
GET Step 3: Get All Owner Profiles
GET Step 4: Get All Balances
GET Step 5: Get All Receipts
POST Step 6: Create KOS
POST Step 7: Create Room
PUT Step 8: Update User Balance
POST Step 9 : Rent Kamar
PUT Step 10: Complete Rent (Admin)
5. DEMO - User Order Flow (Email/Password)
User Order Operations
POST Step 1: Login as User
GET Step 2: Get All KOS
GET Step 3: Get All Rooms
GET Step 4: Get My Balance
POST Step 5: Create Rent (Sewa)
POST Step 6: Create Payment (Buat Tagihan)
PUT Step 7: Process Payment (Bayar)
GET Step 8: Get My Receipts
ENVIRONMENTS
SPECS
FLOWS
GET
{{base_url}} /service3/balance/me
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers 6
Test Results
200 OK
8 ms
390 B
Save Response
JSON
Preview
Visualize
{
"success": true,
"message": "Balance Anda berhasil diambil",
"data": {
"id": 3,
"userId": 3,
"balance": 50000000,
"createdAt": "2026-05-02T14:49:18.000Z",
"updatedAt": "2026-05-02T14:52:23.000Z"
}
}
Connect Git
Console
Terminal
Globals
Vault
Tools
10:20 PM
5/2/2026

ApiGateawayPostman

KOS Management API - Full Collection Copy > 5. DEMO - User Order Flow... > User Order Operations > **Step 5: Create Rent (Sewa)**

POST {{base_url}} /service3/rent

Send

Body: raw, JSON

```
{
    "roomId": 3,
    "startDate": "2024-03-01",
    "endDate": "2024-04-01"
}
```

Body — 201 Created · 14 ms · 473 B

```
{
    "success": true,
    "message": "Rent berhasil dibuat",
    "data": {
        "id": 2,
        "roomId": 3,
        "userId": 3,
        "startDate": "2024-02-29T17:00:00.000Z",
        "endDate": "2024-03-31T17:00:00.000Z",
        "status": "ACTIVE",
        "createdAt": "2026-05-02T14:54:40.000Z",
        "updatedAt": "2026-05-02T14:54:40.000Z"
    }
}
```

10:22 PM 5/2/2026

ApiGateawayPostman

KOS Management API - Full Collection Copy > 5. DEMO - Use... > User Order Operations > **Step 6: Create Payment (Buat Tagihan)**

POST {{base_url}} /service3/payment

Send

Body: raw, JSON

```
{
    "rentId": 2,
    "paymentMethod": "BALANCE"
}
```

Body — 201 Created · 28 ms · 437 B

```
{
    "success": true,
    "message": "Payment berhasil dibuat",
    "data": {
        "id": 1,
        "rentId": 2,
        "totalPrice": 21700000,
        "paymentMethod": "BALANCE",
        "status": "PENDING",
        "createdAt": "2026-05-02T14:55:12.000Z",
        "updatedAt": "2026-05-02T14:55:12.000Z"
    }
}
```

10:22 PM 5/2/2026

ApiGateawayPostman

KOS Management API - Full Collection Copy > 5. DEMO - User Orde... > User Order Operations > **Step 7: Process Payment (Bayar)**

PUT {{base_url}} /service3/payment/1/pay

Send

Query Params

Key	Value	Description

Body — 200 OK · 73 ms · 632 B

```
{
    "success": true,
    "message": "Payment berhasil diproses",
    "data": {
        "id": 1,
        "rentId": 2,
        "totalPrice": 21700000,
        "paymentMethod": "BALANCE",
        "status": "PAID",
        "createdAt": "2026-05-02T14:55:12.000Z",
        "updatedAt": "2026-05-02T14:55:28.000Z",
        "receipt": {
            "id": 1,
            "paymentId": 1,
            "amount": 21700000,
            "description": "Pembayaran sewa room untuk periode 3/1/2024 - 4/1/2024",
            "createdAt": "2026-05-02T14:55:28.000Z",
            "updatedAt": "2026-05-02T14:55:28.000Z"
        }
    }
}
```

10:23 PM 5/2/2026

ApiGateawayPostman

KOS Management API - Full Collection Copy › 5. DEMO - User Order Flow (E... › User Order Operations › **Step 8: Get My Receipts**

GET {{base_url}} /service3/receipt/my

Send

Query Params

Key	Value	Description
Key	Value	Description

Body — 200 OK · 17 ms · 466 B


```
{
    "success": true,
    "message": "Receipt saya berhasil diambil",
    "data": [
        {
            "id": 1,
            "paymentId": 1,
            "amount": 21700000,
            "description": "Pembayaran sewa room untuk periode 3/1/2024 - 4/1/2024",
            "createdAt": "2026-05-02T14:55:28.000Z",
            "updatedAt": "2026-05-02T14:55:28.000Z"
        }
    ]
}
```


ApiGateawayPostman

KOS Management API - Full Collection Copy › 5. DEMO - User Order Flow (Emai... › User Order Operations › **Step 9: Get My Rents**

GET {{base_url}} /service3/rent/my

Send

Query Params

Key	Value	Description
Key	Value	Description

Body — 200 OK · 10 ms · 537 B


```
{
    "success": true,
    "message": "Daftar rent saya berhasil diambil",
    "data": [
        {
            "id": 2,
            "roomId": 3,
            "userId": 3,
            "startDate": "2024-02-29T17:00:00.000Z",
            "endDate": "2024-03-31T17:00:00.000Z",
            "status": "ACTIVE",
            "createdAt": "2026-05-02T14:54:40.000Z",
            "updatedAt": "2026-05-02T14:54:40.000Z"
        }
    ],
    "meta": {
        "page": 1,
```


ApiGateawayPostman

KOS Management API - Full Collection Copy › 5. DEMO - User Order Flow ... › User Order Operations › **Step 10: Get My Payments**

GET {{base_url}} /service3/payment/my

Send

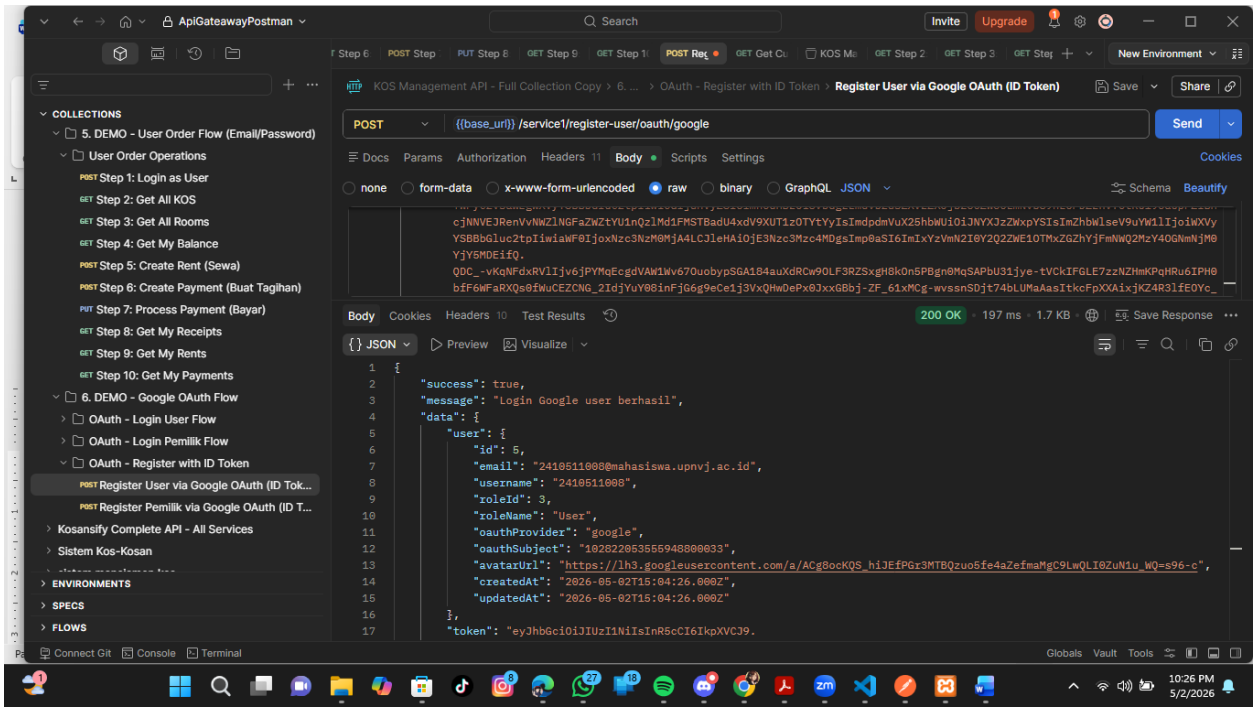
Query Params

Key	Value	Description
Key	Value	Description

Body — 200 OK · 23 ms · 444 B


```
{
    "success": true,
    "message": "Daftar payment saya berhasil diambil",
    "data": [
        {
            "id": 1,
            "rentId": 2,
            "totalPrice": 21700000,
            "paymentMethod": "BALANCE",
            "status": "PAID",
            "createdAt": "2026-05-02T14:55:12.000Z",
            "updatedAt": "2026-05-02T14:55:28.000Z"
        }
    ]
}
```

3) Register id with token



Justifikasi Pemisahan Layanan dalam Arsitektur Microservices

Di backend untuk sistem kos-kosan ini, telah diterapkan pendekatan microservices, yaitu dengan memecah fitur utama menjadi beberapa layanan yang berdiri
sendiri seperti Layanan Auth, Layanan Pengguna, dan Layanan Kos. Pemisahan ini dilakukan agar setiap fungsi memiliki tanggung jawab yang jelas dan memudahkan pengembangan sistem.

1. Auth Service Dipisahkan

Auth Service menangani proses autentikasi dan keamanan, seperti login, JWT, refresh token, dan Google OAuth.

Alasan dipisahkan:

a. Sistem autentikasi memiliki tingkat keamanan tinggi dan logika khusus.
b. Dapat digunakan oleh berbagai layanan lain tanpa duplikasi kode.
c. Jika terjadi perubahan metode login, tidak memengaruhi service lain.

Jika monolitik:

a. Seluruh proses autentikasi bercampur dengan logika bisnis lain.
b. Risiko bug keamanan lebih besar karena semua modul saling terhubung.

c. Perubahan login dapat memengaruhi keseluruhan aplikasi.

2. User Service Dipisahkan

User Service bertugas mengelola data profil pengguna, role, dan informasi owner.

Alasan dipisahkan:

a. Data pengguna memiliki lifecycle berbeda dengan data kos.
b. Memudahkan pengelolaan role dan profil secara terstruktur.
c. Skalabilitas lebih baik jika jumlah pengguna meningkat.

Jika monolitik:

a. Modul user bercampur dengan transaksi kos dan pembayaran.
b. Query database menjadi kompleks karena semua fitur berada dalam satu sistem besar.
c. Maintenance menjadi lebih sulit.

3. Kos Service Dipisahkan

Kos Service mengelola data kos, kamar, fasilitas, pembayaran, dan penyewaan.

Alasan dipisahkan:

a. Memiliki proses bisnis yang cukup besar dan kompleks.
b. Operasi CRUD kos dan transaksi lebih fokus.
c. Dapat dikembangkan secara independen tanpa mengganggu autentikasi atau user.

Microservices berbeda dengan arsitektur monolitik karena sistem dibagi menjadi beberapa layanan sesuai fungsi masing-masing, sedangkan monolitik menggabungkan seluruh fitur dalam satu aplikasi. Dari sisi skalabilitas, microservices lebih fleksibel karena setiap service dapat ditingkatkan kapasitasnya secara terpisah, sementara monolitik harus meningkatkan seluruh sistem sekaligus. Dalam maintenance, microservices lebih mudah dikelola karena struktur modul terpisah, sedangkan monolitik menjadi lebih rumit ketika aplikasi semakin besar. Selain itu, deployment pada microservices dapat dilakukan per service tanpa memengaruhi sistem lain, sedangkan monolitik mengharuskan deployment seluruh aplikasi. Dari segi risiko, error pada microservices cenderung terisolasi, sementara pada monolitik kesalahan pada satu bagian dapat berdampak ke seluruh sistem.

Kesimpulan

Pemisahan service dilakukan agar sistem memiliki separation of concern, lebih mudah dikembangkan, lebih aman, dan scalable. Pendekatan microservices lebih cocok untuk sistem kos-kosan karena memiliki beberapa domain berbeda seperti autentikasi, manajemen user, dan pengelolaan kos yang dapat berjalan secara independen.